

Learning Objective 3.1: Describe an aperture distribution, compute the beamwidth, sidelobe level and aperture efficiency it produces, and use those results to size an aperture for a specification.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **The uniform aperture.** A rectangular horn is fed so that the field across its 24 cm broad dimension is essentially constant in amplitude and phase. It operates at 12 GHz.

- (a) In one or two sentences, state the relationship between the aperture distribution and the far-field pattern, and say what the space frequency $u = (L/\lambda)\sin(\theta)$ does for you that the angle θ does not.

- (b) Find the aperture length in wavelengths and the half-power beamwidth in the broad-dimension plane.

$\theta_{\text{HP}} =$

- (c) Find the angle of the first null measured from broad-side.

$\theta_{\text{null}} =$

- (d) State the first sidelobe level of this aperture, and explain in one sentence why lengthening the horn would not change it.

2. **Aperture efficiency.** Aperture efficiency converts physical area into gain through $G = \eta_{\text{ap}} 4\pi A/\lambda^2$.

- (a) Write the definition of η_{ap} in terms of the aperture field, and state in words what the numerator and the denominator each represent.

- (b) A line aperture carries a triangular illumination, $E_a = 1 - 2|x|/L$. Evaluate the two integrals over $\xi = x/L$ and compute η_{ap} .

$\eta_{\text{ap}} =$

- (c) A pyramidal horn has an $8 \text{ cm} \times 6 \text{ cm}$ mouth and a measured aperture efficiency of 0.51 at 10 GHz. Find its gain in dBi.

$G =$

- (d) Find the effective aperture A_e of that horn, and say in one sentence what the difference between A_e and the mouth area physically represents.

$A_e =$

3. **The taper trade.** A line aperture is 40λ long. The canonical taper table is reproduced here.

Illumination	First sidelobe	HPBW ($\times \lambda/L$)	η_{ap}
Uniform	−13.3 dB	0.886	1.00
Cosine	−23 dB	1.19	0.81
Triangular	−26.5 dB	1.27	0.75
Cosine ²	−31.5 dB	1.44	0.667

- (a) Find the half-power beamwidth with uniform illumination.

$$\theta_{\text{HP}} =$$

- (b) The system requires a first sidelobe no higher than −25 dB. Choose the illumination from the table that meets it with the narrowest beam, and find the beamwidth you get.

$$\theta_{\text{HP}} =$$

- (c) Find the gain penalty in dB that this taper costs relative to uniform illumination on the same aperture.

$$\Delta G =$$

- (d) Suppose the 1.27° beamwidth is also a requirement. How long must the triangular-illuminated aperture be, and what has the sidelobe requirement actually cost you? Answer with a number and one sentence.

$$L' =$$

4. **Scaling a reflector.** A circular reflector of diameter $D = 1.2$ m is uniformly illuminated to a good approximation and has an aperture efficiency of 0.60 at 10 GHz. Use $\theta_{\text{HP}} = 58.4^\circ \lambda/D$.

(a) Find the half-power beamwidth at 10 GHz.

$$\theta_{\text{HP}} = \boxed{}$$

(b) Find the gain in dBi at 10 GHz.

$$G = \boxed{}$$

(c) The same dish is used at 30 GHz with the same aperture efficiency. Find the new beamwidth and the new gain.

$$\theta_{\text{HP}} = \boxed{}$$

$$G = \boxed{}$$

(d) The dish did not change, yet the gain rose by 9.5 dB. Explain where that gain came from, and name one practical reason a real reflector will fall short of the predicted 49.3 dBi at 30 GHz.

5. **Reading a measured pattern.** A student measures the pattern of an unknown aperture antenna at 10 GHz on the course range. The main lobe is symmetric about boresight, the first nulls fall at $\pm 3.4^\circ$, and the first sidelobes read 19 dB below the peak.

- (a) Assuming the illumination is uniform, find the aperture dimension in the measured plane.

 $L =$

- (b) Predict the half-power beamwidth that this aperture should show.

 $\theta_{\text{HP}} =$

- (c) The measured sidelobe is 19 dB down, not 13.3 dB down. Give two distinct explanations, one about the antenna and one about the measurement.

- (d) The same antenna is later found to have $\eta_{\text{ap}} = 0.44$, although its amplitude taper alone would predict about 0.81. Name two mechanisms that account for the difference, and state whether each one belongs to aperture efficiency or to radiation efficiency.

Documentation: _____